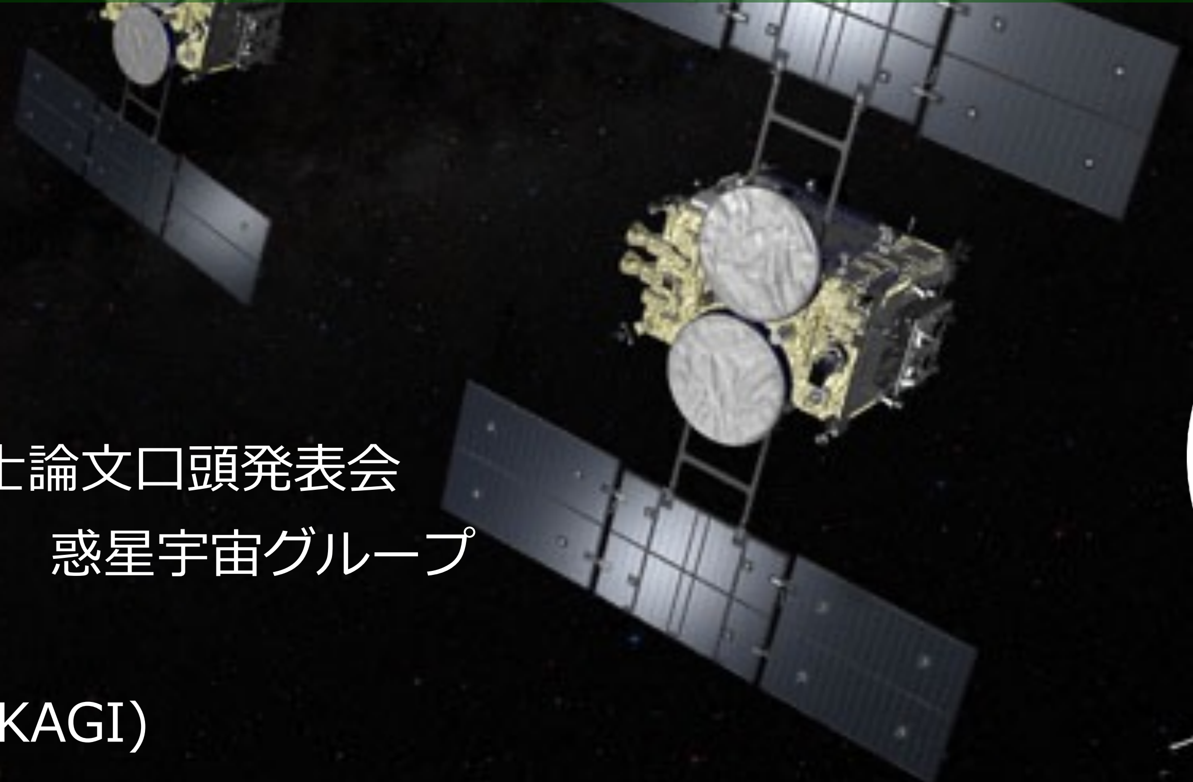
せいめい望遠鏡を用いた３色同時測光観測によるはやぶさ２#ターゲット小惑星“Torifune”の表面カラー特性

Surface Color Properties of the Hayabusa2# Target Asteroid “Torifune” from Simultaneous Three-Filter Photometric Observations with the Seimei Telescope



2026.02.03 11:15-11:30 修士論文口頭発表会

北海道大学理学院宇宙理学専攻　惑星宇宙グループ

土井知也 (Kazuya DOI)

指導教員：髙木聖子 (Seiko TAKAGI)

0. Contents of my Thesis

Submitted to PASJ

Research 1

ピリカ望遠鏡を用いた偏光観測による月軌道以下まで接近した小惑星 ”2024 MK” の表面散乱特性

Polarimetric Observations of the Near-Earth Asteroid 2024 MK During its close approach within Lunar Distance

Today’s talk

Research 2

せいめい望遠鏡を用いた３色同時測光観測によるはやぶさ2# ターゲット小惑星 “Torifune” の表面カラー特性

Surface Color Properties of the Hayabusa2# Target Asteroid “Torifune” from Simultaneous Three-Filter Photometric Observations with the Seimei Telescope

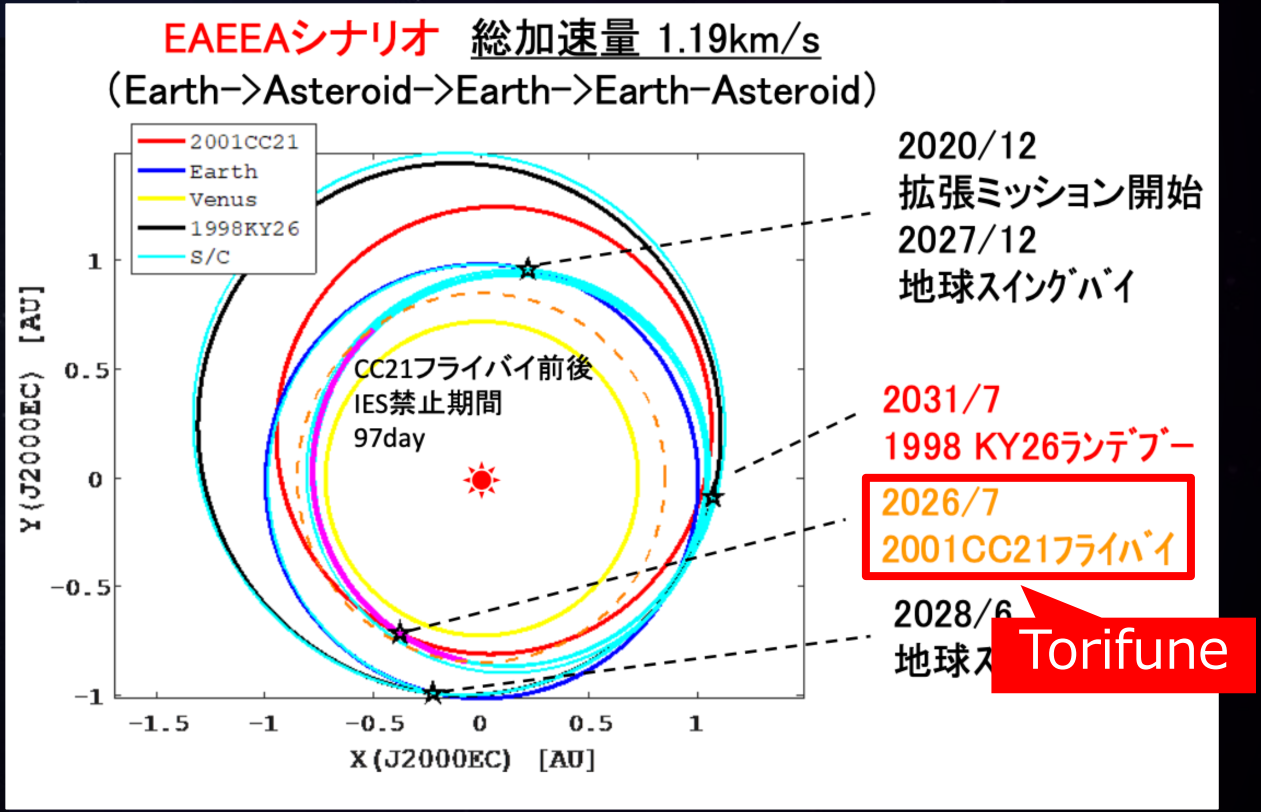
1. Hayabusa2# (SHARP)

Small **H**azardous **A**steroid **R**econnaissance **P**robe

【Purpose of this mission】

1. Evolution of long-term deep space cruise technology
2. Exploration of fast rotator asteroid
3. Acquisition of science and technology for planetary defense

(Tsuda+, 2025)



【Targets】

1. **Torifune** (2001 CC21)
   Flyby in July 5, 2026
2. 1998 KY26
   Rendezvous in July 2031

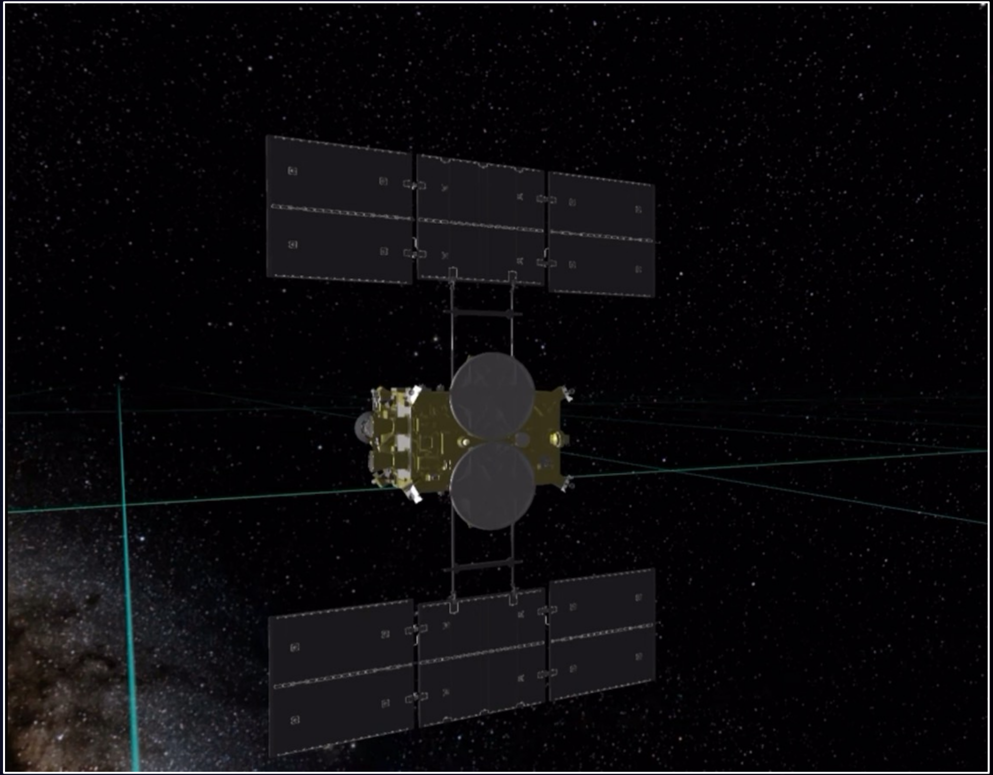
1. Super-close High-speed Flyby

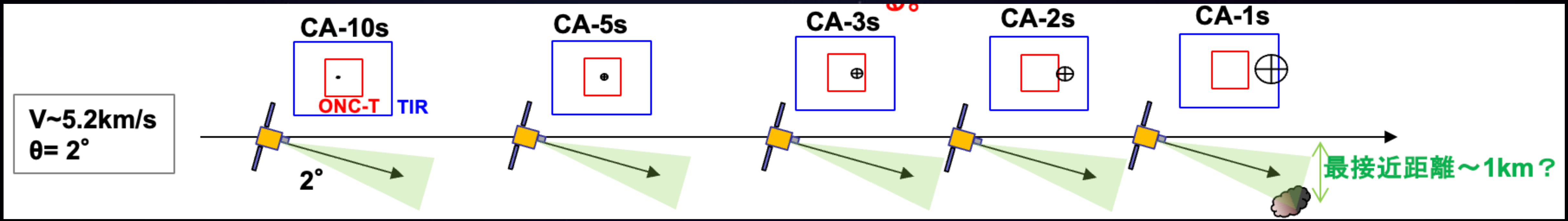
- **Closest Approach (CA) distance**: **~<u>1 km</u>** (TBD by JAXA)
- **Relative velocity**: ~**<u>5 km/s</u>**

→ Designed for the Ryugu rendezvous, can't focus, must close approach

→ Exposure time & gain must be **<u>preset</u>**

→ Only **<u>one side</u>** of Torifune can be observed during the flyby

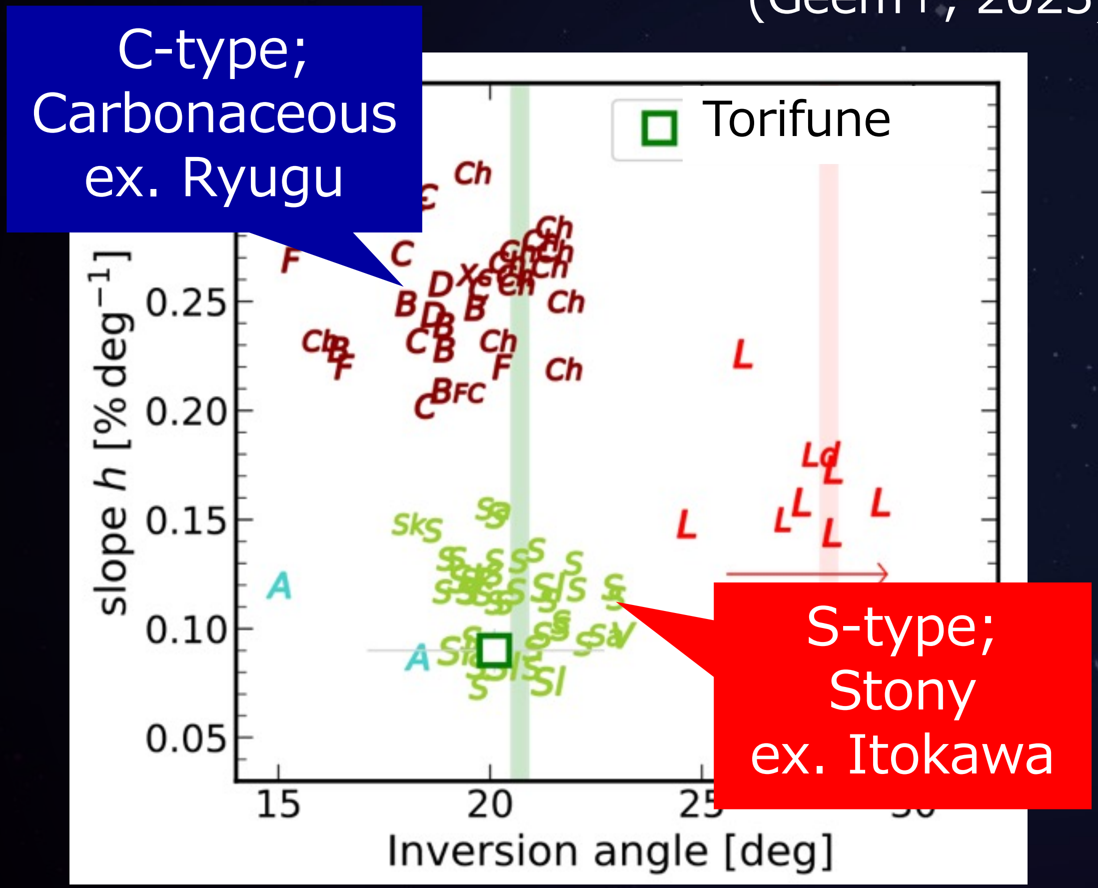




2. Previous Observations

【Polarimetric Observation】

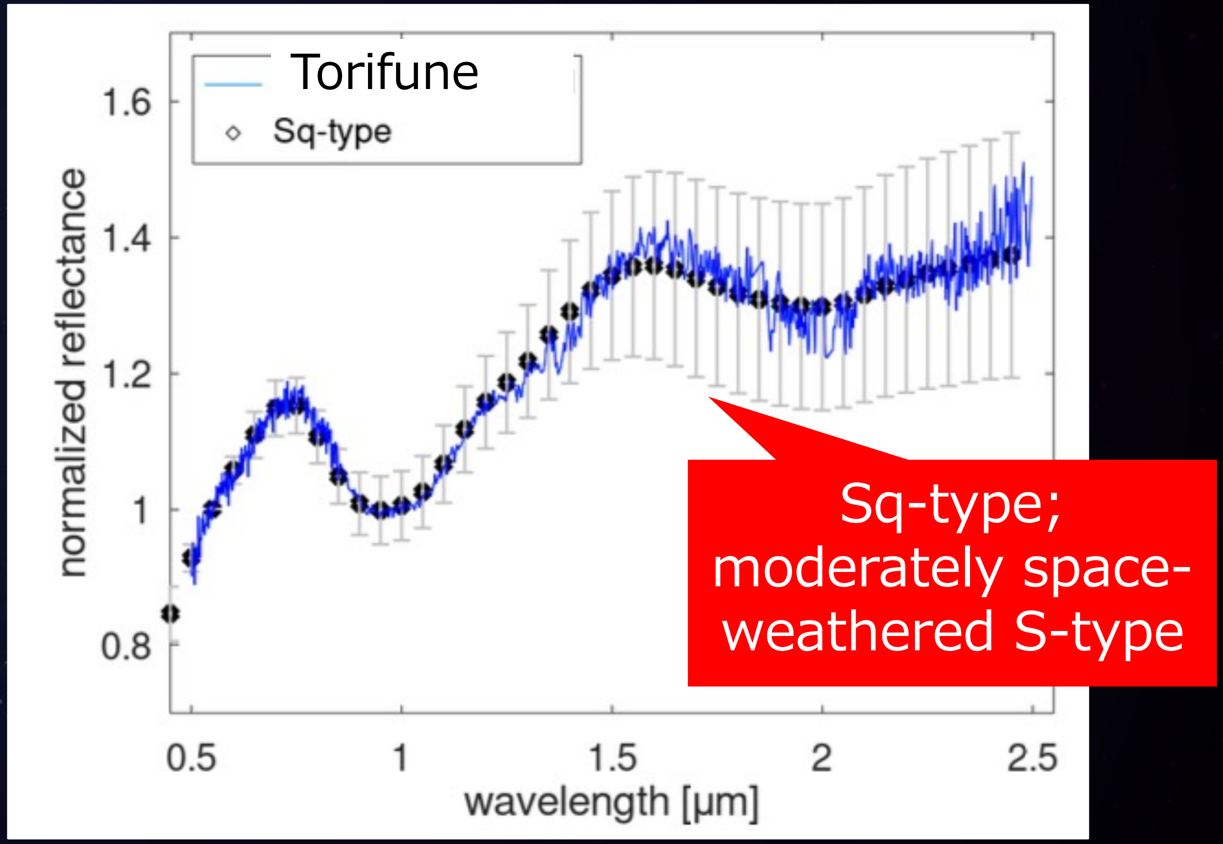
(Geem+, 2023)



S-type Polarimetric properties

【Spectroscopic Observation】

(Geem+, 2023; Popescu+, 2025)



10.2 m GTC/OSIRIS & 3.2 m IRTF/SpeX

Sq- (S-)type Reflectance spectrum

2. Previous Observations

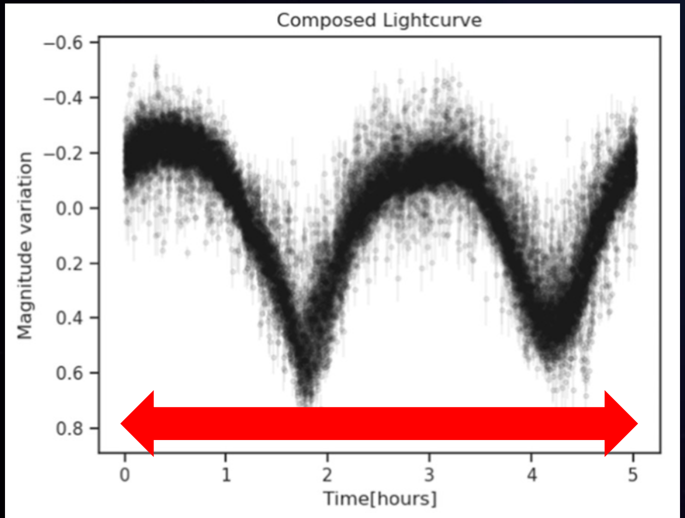
【Photometric Observation】

(Popescu+, 2025)

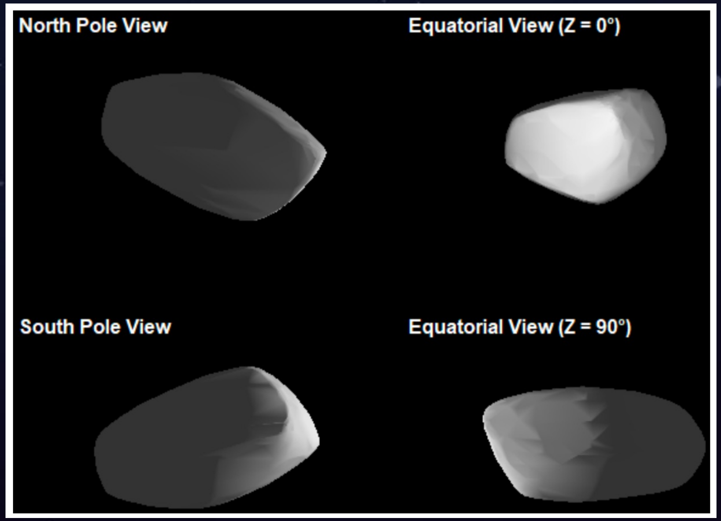
- **S-type** color-index (g-r, r-i, i-z_s)
- Small variation, **Surface homogeneity**

【Light-curve Observation】

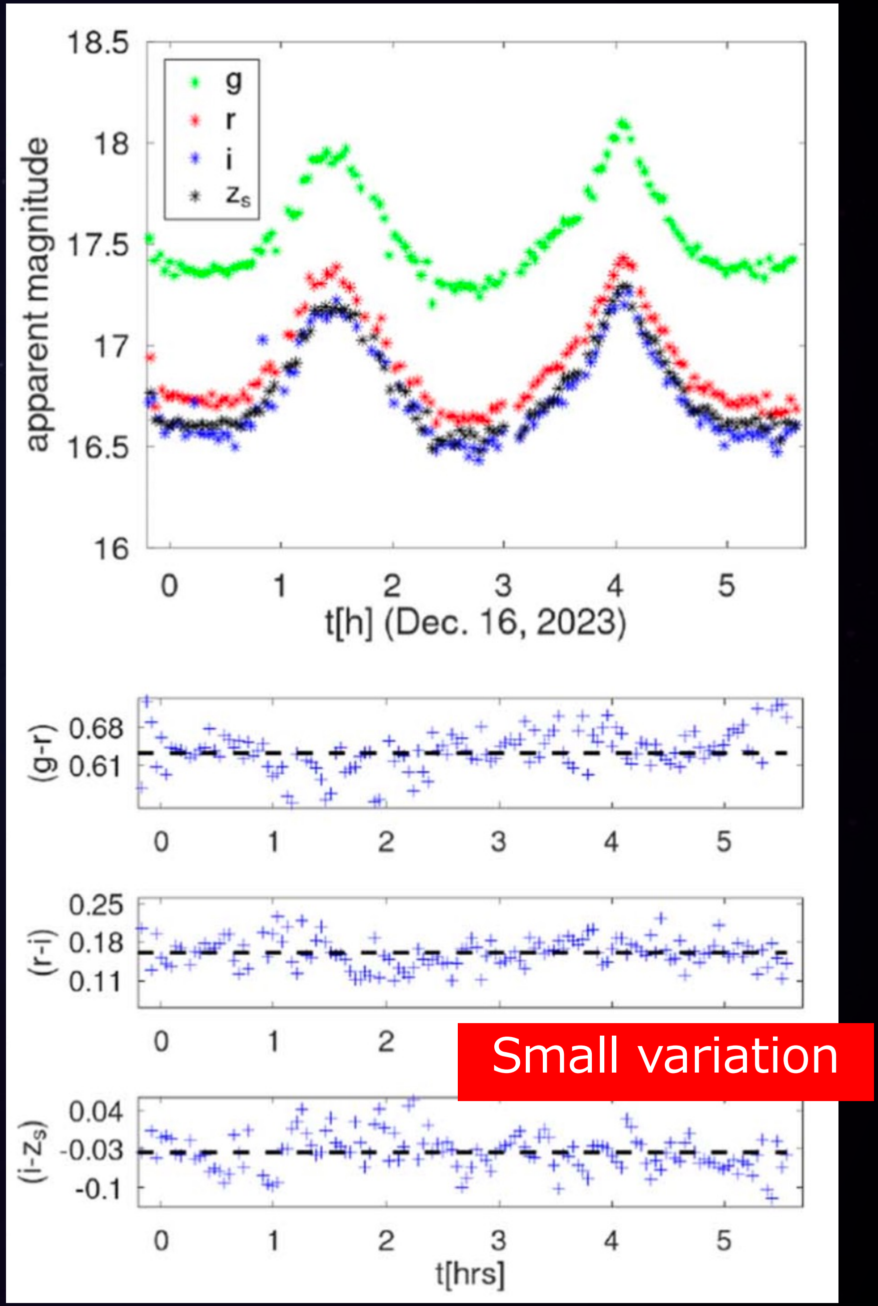
(Popescu+, 2025; Fatka+, 2025 etc.)



Rotation period: **~5.02 h**



3D model, ~500 m



3. Objectives

【Significance of observations before spacecraft missions】

Technical: Pre-setting the exposure time and gain of onboard cameras

→ can't be properly determined unless the surface color **in advance**

Scientific: Surface color properties of Torifune

→ **surface homogeneity or heterogeneity**

→ **highest scientific side** for observation

→ whether the one side information **represents Torifune's entire surface**

【Objectives】

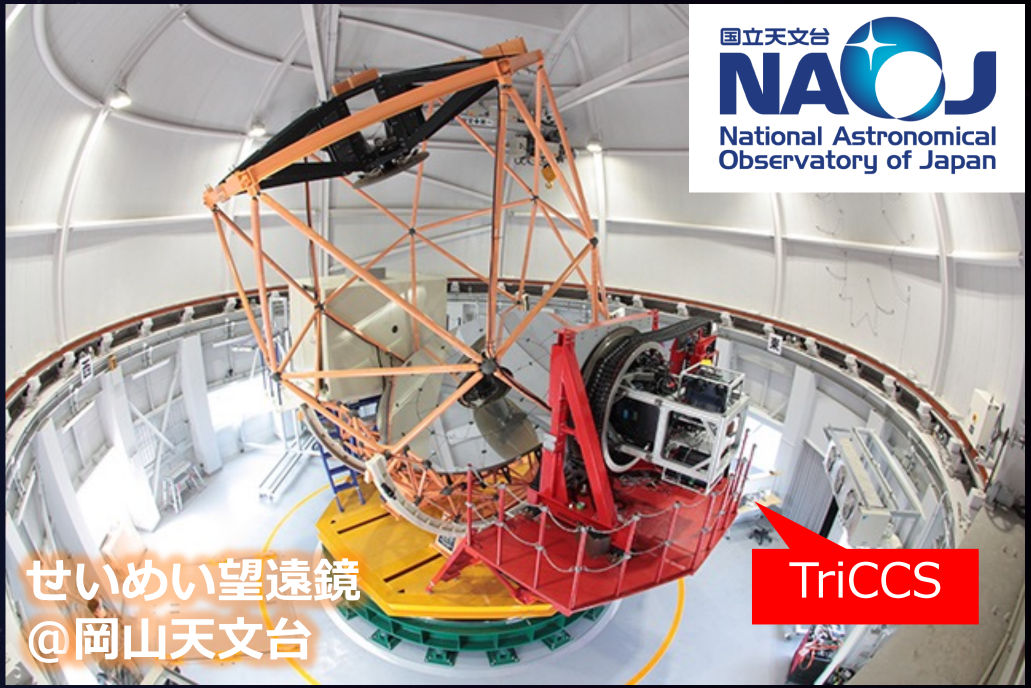
Simultaneous three-filter photometry of Torifune before the Hayabusa2# flyby to derive its **surface color properties** with **high time resolution and high photometric accuracy**

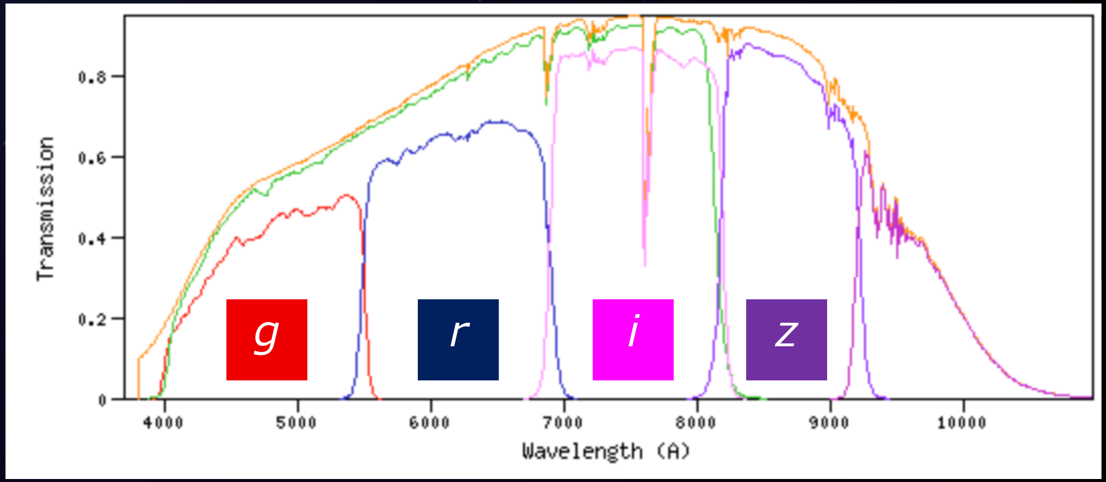
4. 3.8 m Seimei Telescope/TriCCS

- Proposal was approved by NAOJ
- The largest optical telescope in East Asia, optimal for observing faint asteroids
- Simultaneous three-filter photometry in Pan-STARRS system *g*, *r*, *i* or *z*-filter

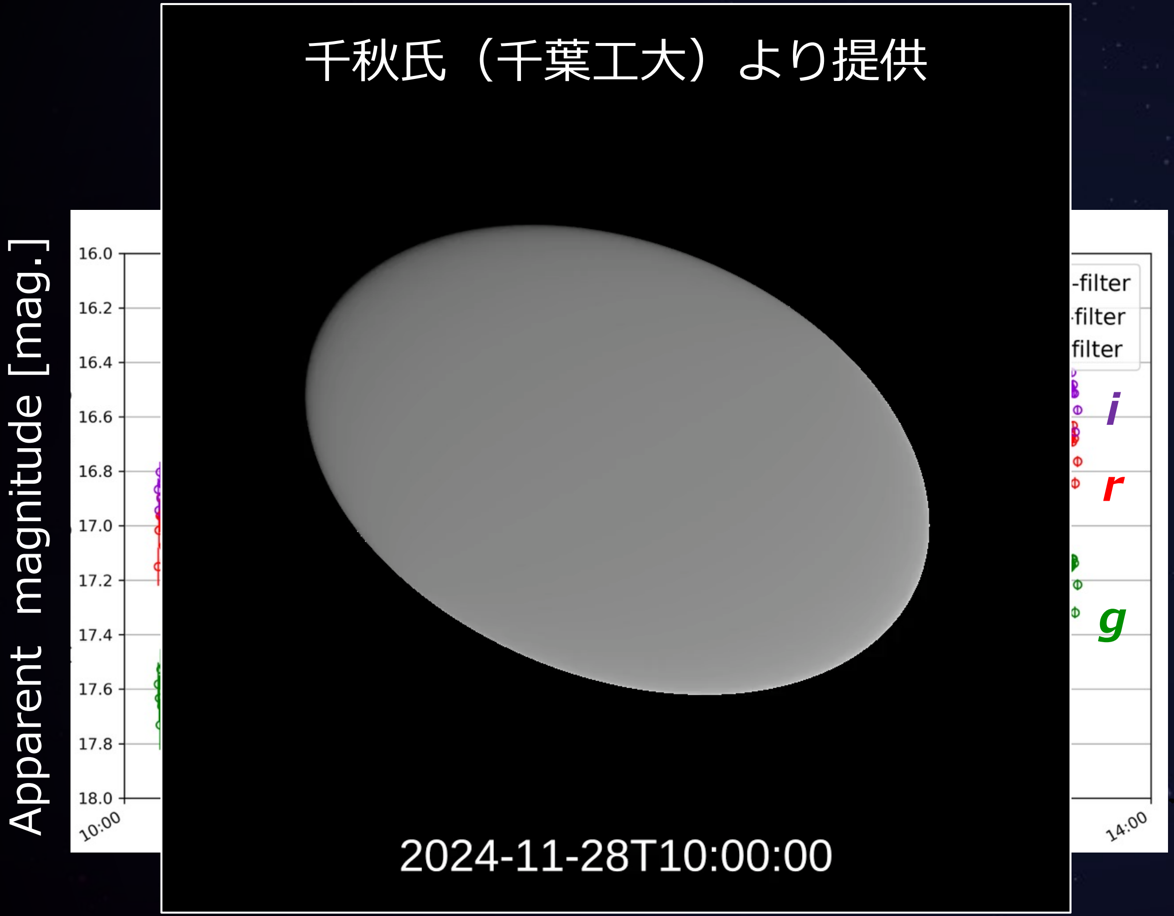
Tab. Observational list

Data	Phase angle	Filter	Notes
2024 Nov. 27	6°	*g*, *r*, *i*	Cloudy
2024 Nov. 28	7°	*g*, *r*, *i*	Partly Cloudy
2025 Oct. 22	20°	*g*, *r*, *i* & *g*, *r*, *z*	Partly Cloudy
2025 Nov. 6	25°	*g*, *r*, *i*	Sunny DDT



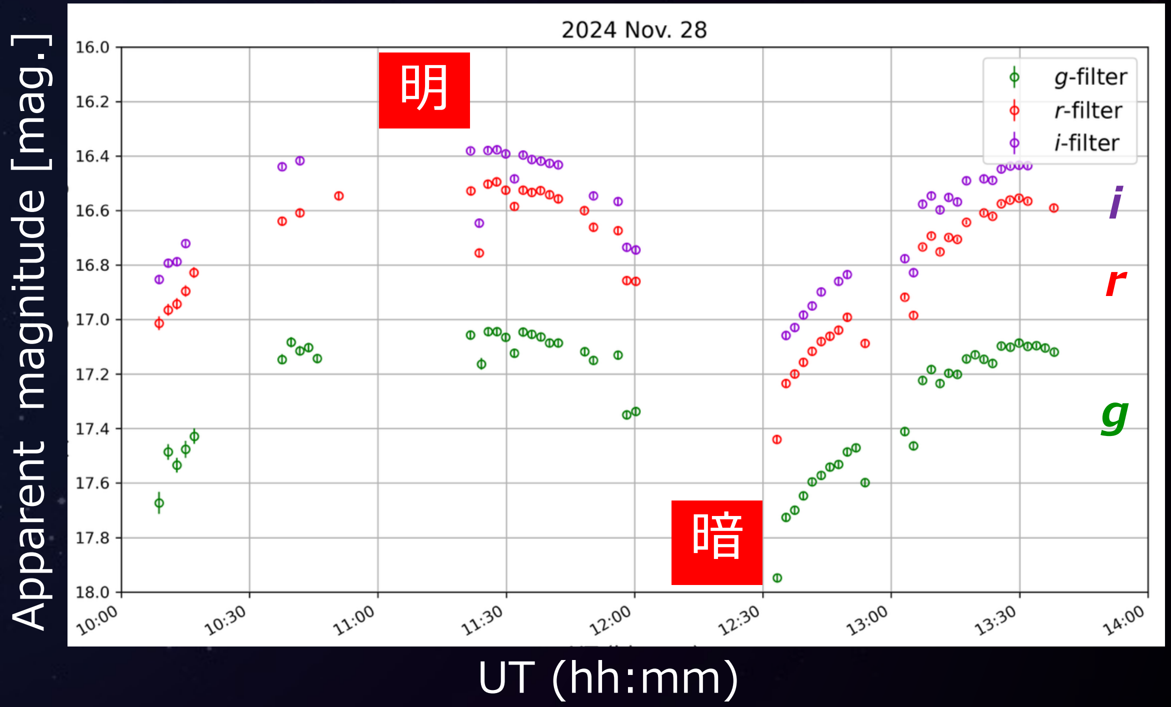


5. Results: Multi-Color Light-curve



(ex. 2024. Nov. 28)

高測光精度（120 s, S/N > 100）



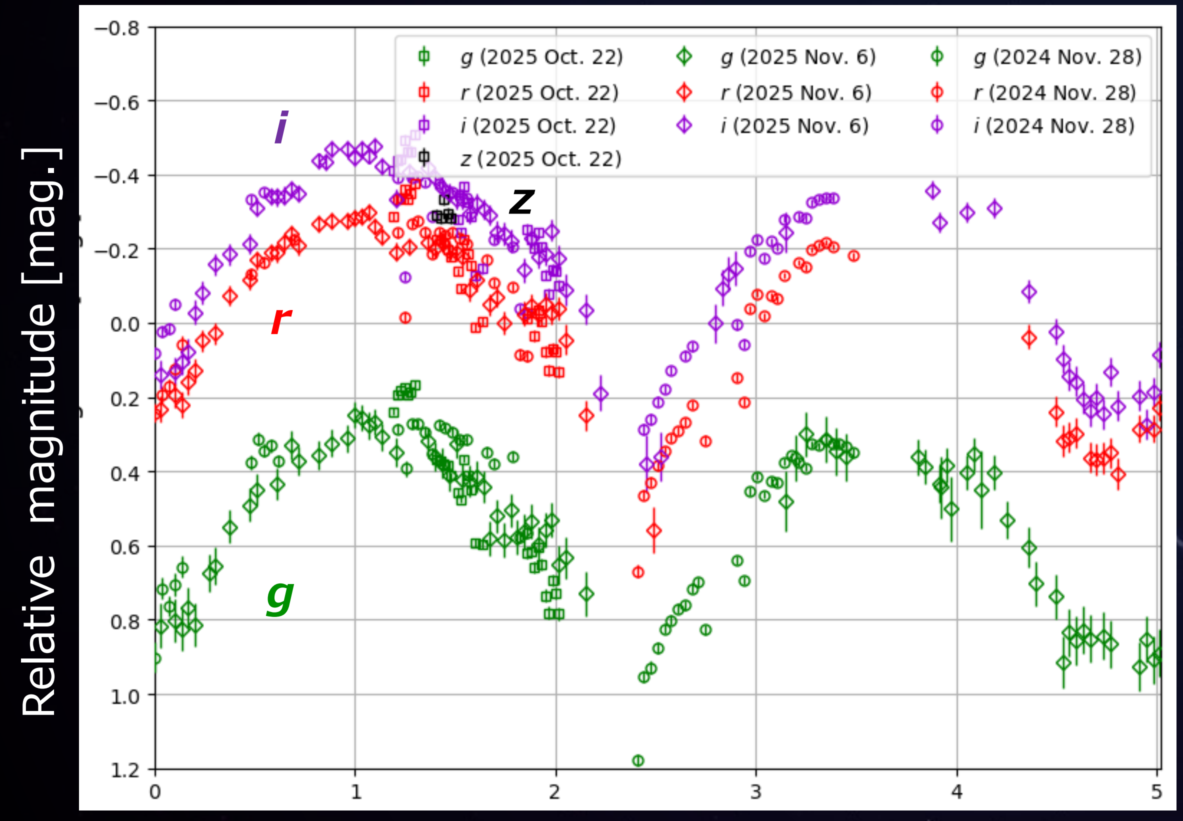
- **High time resolution** & **high photometric accuracy** data
- Consistent with the observational geometry model

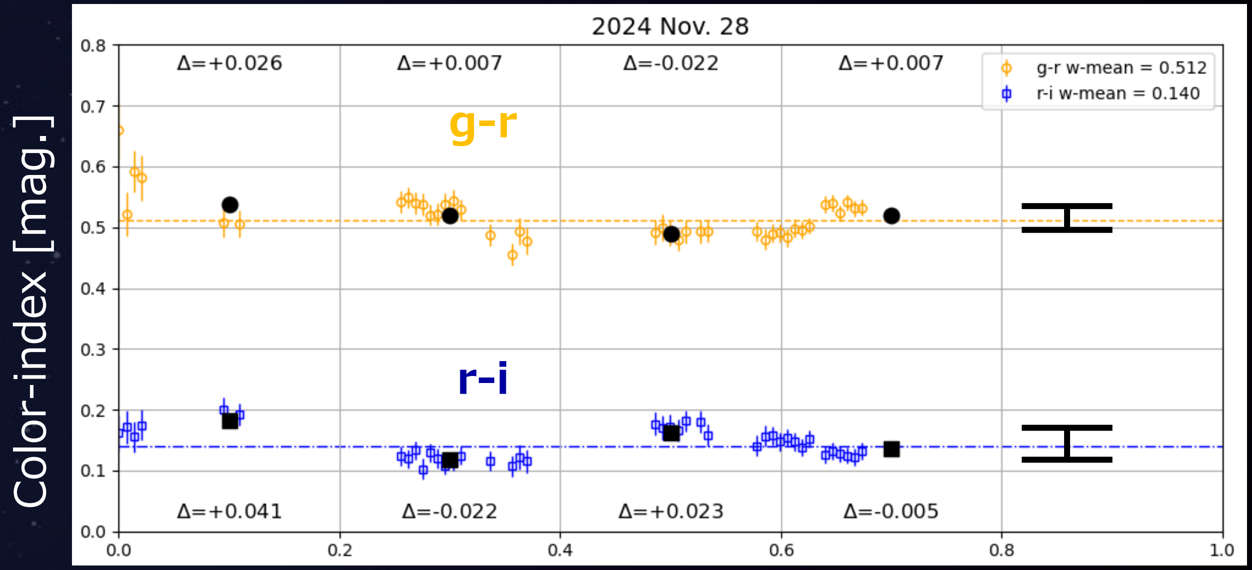
5. Results: Color-index variation

【Multi-color Light curve】

(ex. 2024. Nov. 28)

Fold 3 nights data with 5.02 h





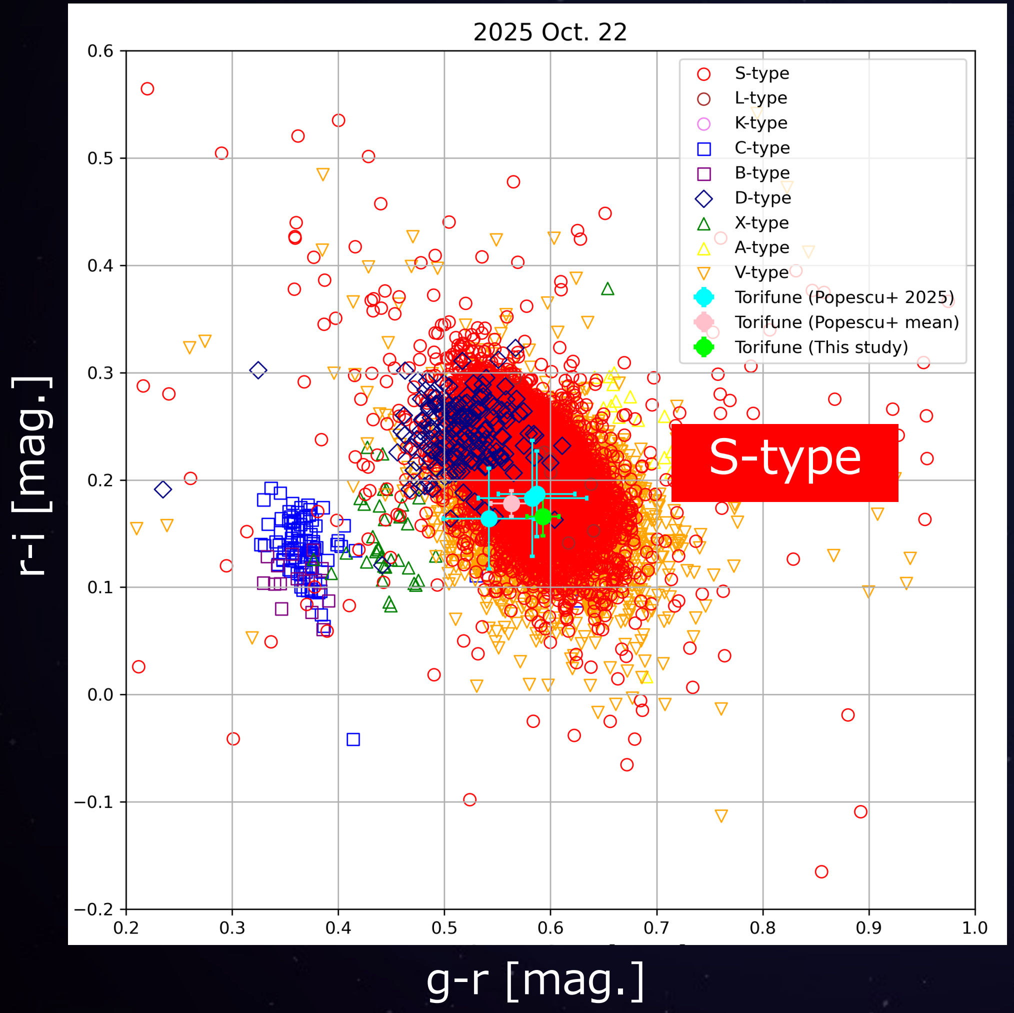
Δg-r, Δr-i **<u>~0.05 mag.</u>** < 0.10 mag.

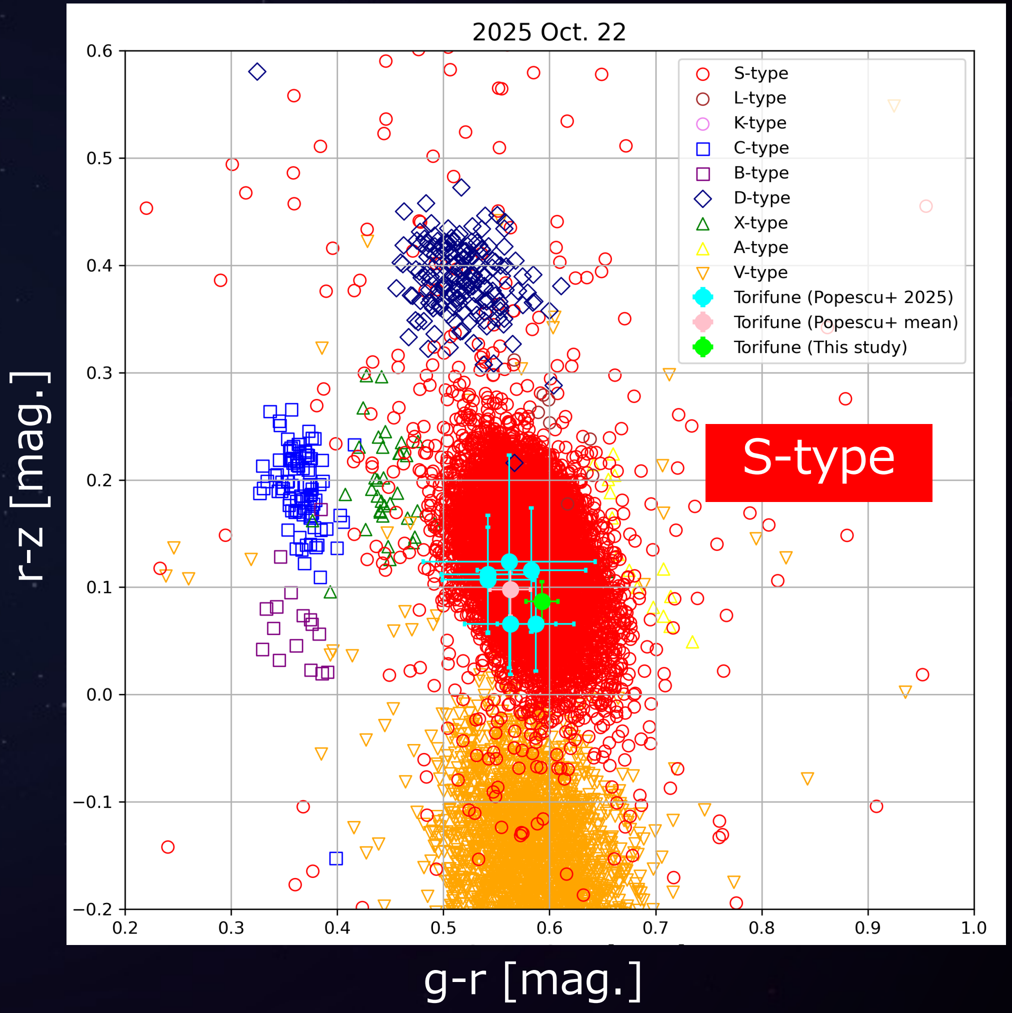
→ **Small color variation**

→ **<u>Globally homogeneous surface color properties</u>**

5. Results: Mean Color-index

(ex. 2025. Oct. 22)





- **S-type color-index (g-r, r-i, r-z)**

5. Results: Reflectance spectrum

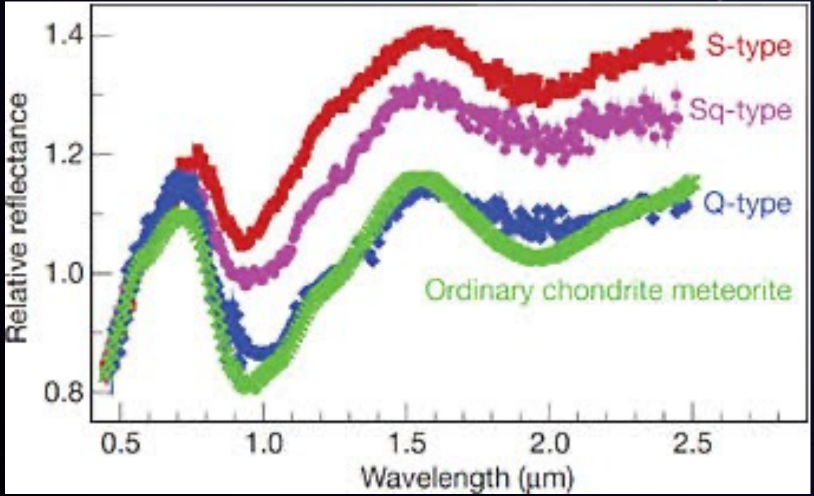
(ex. 2025. Oct. 22)

- Color-index (g-r, r-i, r-z) transform Reflectance;

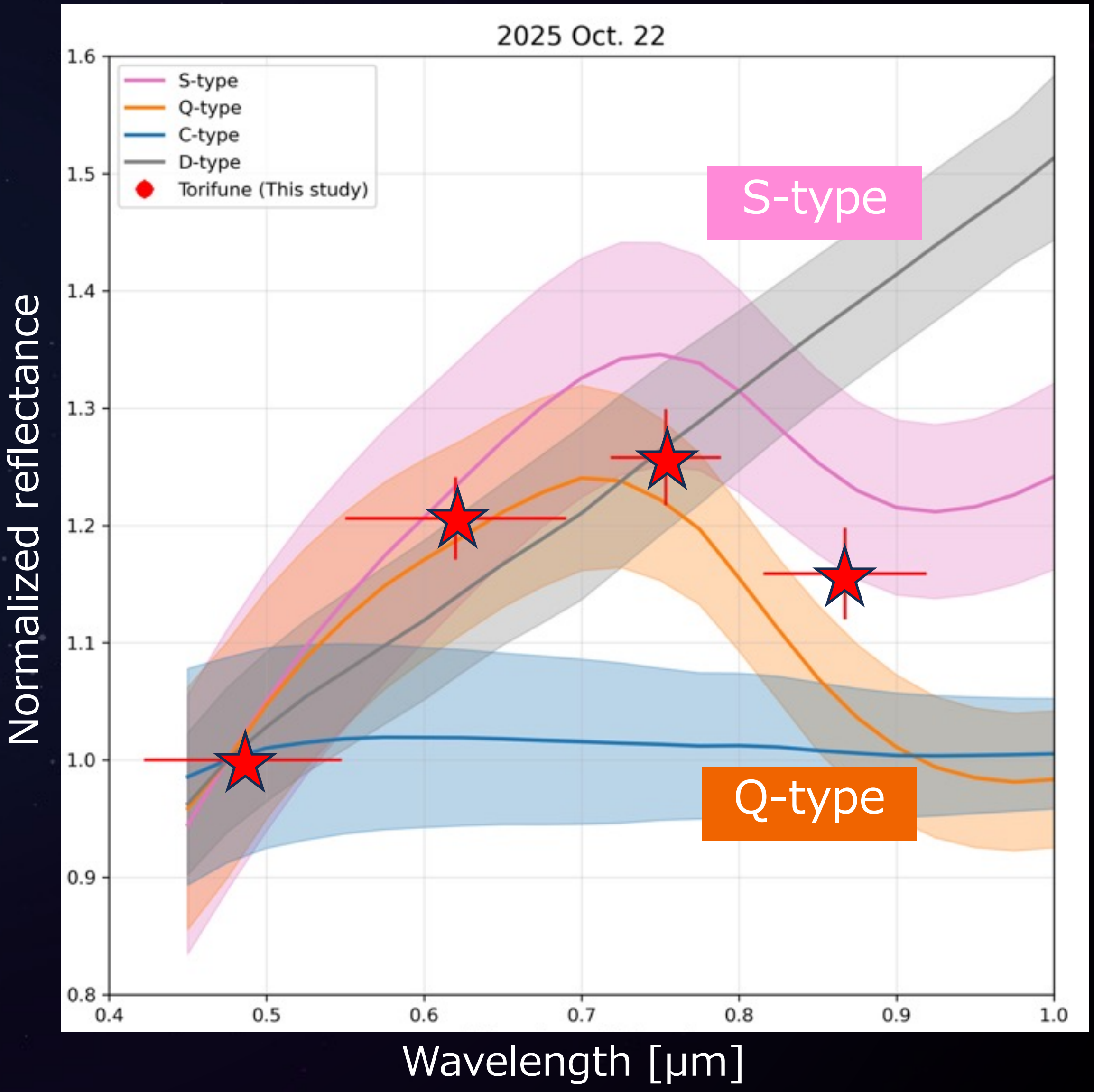
$$R_r = 10^{-0.4[(r-g)_{obj}-(r-g)_{sun}]}$$
$$R_i = 10^{-0.4[(i-g)_{obj}-(i-g)_{sun}]}$$
$$R_z = 10^{-0.4[(z-g)_{obj}-(z-g)_{sun}]}$$

- Between **S-type** and **Q-type**

→ **Sq-type color properties**

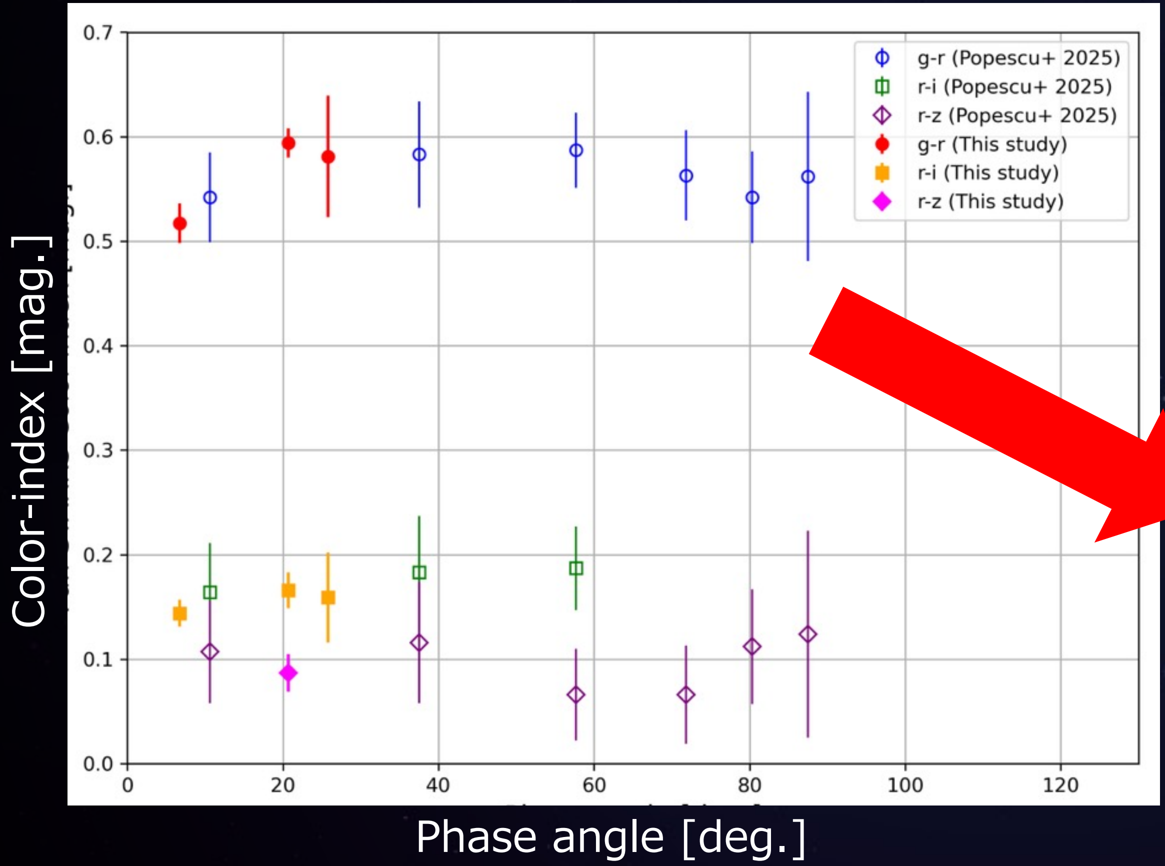


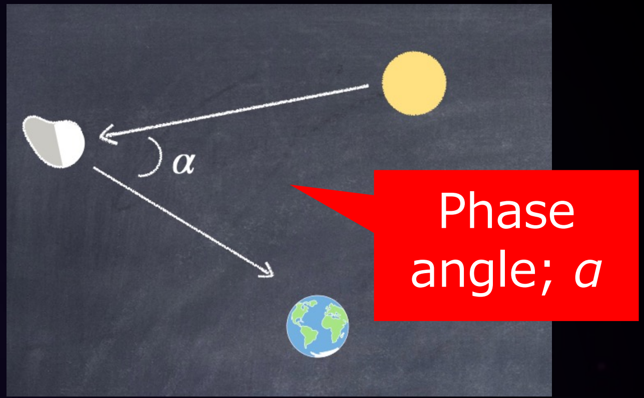
S-, Sq-, Q-type reflectance spectrum (Binzel+, 2010)



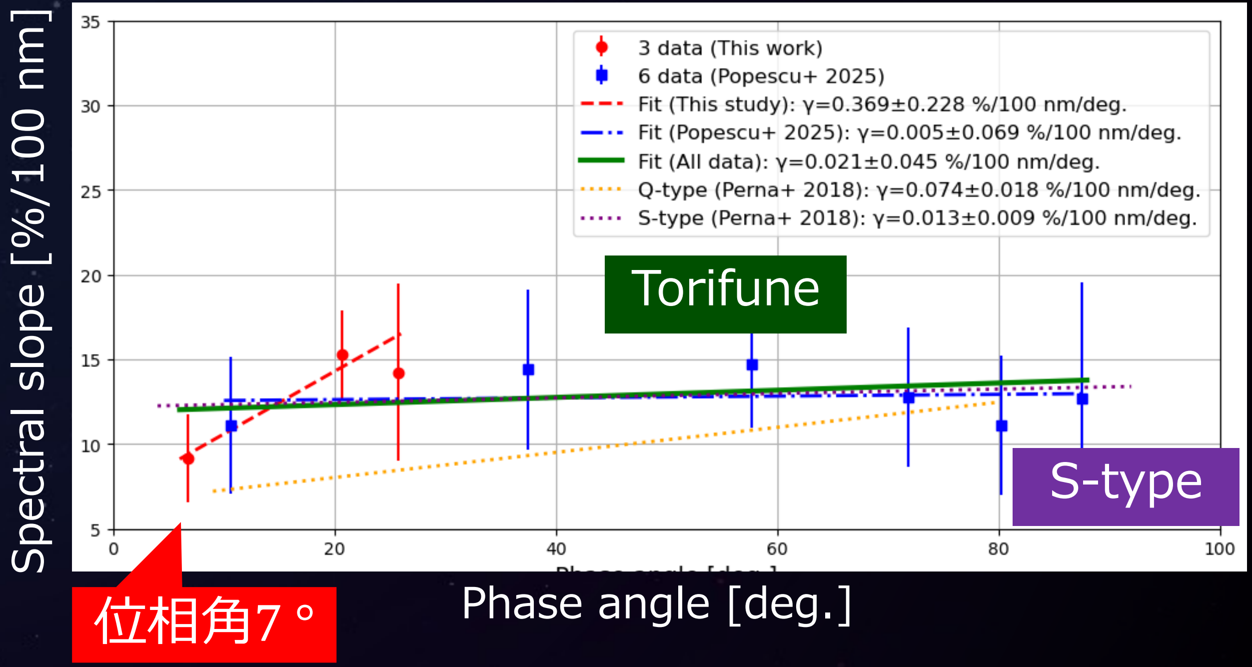
5. Results: Phase reddening

各位相角での色指数





スペクトル（R_g ~ R_r）の傾き



- **S-type phase reddening**
- Flyby observations at **low phase angle** (behind the Sun)

6. Summary

- せいめい望遠鏡を用いて、3色同時測光観測を実施した
- 先行研究よりも**高時間分解能**、**高測光精度**のカラーを取得した

- **全面的に均一**なカラー特性を持つ
- **S-type**、特に**Sq-type**のカラー特性を持つ

- **S-typeのPhase reddeningの傾向**を持つ
- 太陽を背にしたフライバイ観測において重要な**低位相角でのカラー**を取得した

- PSJにて投稿、はやぶさ2＃特集号にエントリー予定